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Cuffless blood pressure estimation from PPG signals and its derivatives using deep learning models

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ABSTRACT

Blood pressure (BP) is a direct indicator for hypertension, therefore, continuous and non-invasive BP monitoring is essential for reducing future health complications. Most non-invasive blood pressure monitors use the oscillometric technique, which can be cumbersome and impractical. To address this problem, we explore several features extracted from the Photoplethysmogram (PPG) waveform and its first and second derivatives and employ deep learning recurrent models for non-invasive cuffless estimation for systolic and diastolic BP. In this research, three techniques have been considered including statistical and machine learning techniques for reducing the collinearity and redundancy in the input feature vector. The estimation models consist of a one bidirectional recurrent layer, followed by a series of stacked conventional recurrent layers and an attention layer. All models were evaluated on 942 subjects collected from the MIMIC II dataset. The best performing model (consists of one bidirectional layer followed by several Long Short-Term Memory layers and an attention layer) achieved a mean absolute error, and standard deviation of 4.51 ± 7.81 mmHg for systolic BP (SBP), and 2.6 ± 4.41 mmHg for diastolic BP (DBP). The results show that the deep learning model trained on features extracted from one PPG sensor yield good performance for continuous and cuffless BP monitoring. Additionally, the results fulfil the international standard for cuffless BP estimation.

1. Introduction

Blood pressure is the force exerted by the blood flow against the walls of the blood vessels. BP is a significant physiological indicator of human health that provides fundamental information for physicians. Moreover, BP is closely related to the function of the heart as well as the thickness and elasticity of the blood vessels [1]. BP is a primary vital sign and is used widely in clinical practice for assessing the cardiovascular functions of a patient. As the heart beats, blood flows from the heart to the peripheral vasculature causing the BP to vary between systolic blood pressure (SBP) and diastolic blood pressure (DBP). SBP refers to the maximum pressure in the blood vessels, as a result of the rapid flow of blood, caused by the systolic contraction of the heart, whereas DBP refers to the low pressure, in the relaxed blood vessels, when the heart rests between beats.

High BP or hypertension increases the risk of having health problems [2]. High BP adds an excessive amount of force against the blood arteries. Over time, this strain may cause arteries to become less flexible with narrower interior space (lumen), which in turn increases the chances of having a clot. This leads to various pathologies relating to cardiovascular disease (CVD) such as stroke and heart attack. According to the World Health Organisation (WHO), CVDs cause the highest number of mortalities worldwide [3]. Additionally, more than one billion of the global population are affected by hypertension, which is known as a major contributor to CVD [4]. This makes hypertension a global public health issue. Therefore, continuous BP measurement is integral for the early detection, diagnosis and treatment of CVDs.

Currently, BP measurement techniques can be divided into direct or indirect methods. The direct method is known as catheterisation [5]. It provides continuous BP values; however, it is invasive, painful and may cause infection. Also, such a technique is only applicable to patients undergoing surgical interventions. The most commonly used BP measurement techniques in health-care are based on conventional cuff-type devices [6]. The cuff-based measurements can provide SBP and DBP values for a given point in time noninvasively. However, despite this advantage, cuff-based BP measurement is mostly intermittent, tedious, and inconvenient. Moreover, it may result in inaccurate BP values, which can lead to misdiagnosis, such as the "masked hypertension" [7]

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or the “white coat hypertension” [8]. The former scenario occurs when a normal BP is observed for a patient when their actual BP is high, as opposed to the latter where high BP is observed for a nervous or anxious patient with actually a normal BP. Thus, cuff-based devices are not practical for the early detection and diagnosis of hypertension. Therefore, there is an unmet need for a new non-invasive, continuous, cuffless, and accurate self-measuring BP method.

In order to overcome the aforementioned limitations, over the past two decades, extensive research have been conducted for developing robust non-invasive cuffless and continuous BP measurement methods. Consequently, researchers proposed physiological signals as possible alternatives for obtaining cuffless and continuous BP estimation. The most common approaches are mainly based on Pulse Wave Velocity (PWV) [9] and Photoplethysmogram (PPG) morphological features [10–14]. The former is based on the arterial wave propagation theory and is derived using pulse transit time (PTT) through the following equation:

$$PWV = \frac{D}{PTT}$$

Where D is the distance between two sensors (ECG and PPG sensor or two PPG sensors) placed on the body and PTT is the time that takes for the blood pulse to travel between these two arterial sites [15]. PTT is negatively correlated with BP and can be used for cuffless non-invasive BP monitoring [16]. Several studies have attempted to approximate BP through PTT [17–19], however, it should be noted that the majority of these studies have used pulse arrival time (PAT) [20–22], which was referred to as PTT interchangeably in the literature. PAT is easier to measure and refers to the time interval for the blood pulse to travel between the aortic valve opening and the arrival of the pulse wave to a peripheral point on the body such as fingertip, toe, earlobe, etc. PAT is defined as the time delay between the R peak of the electrocardiogram (ECG) and a point on the rising edge of a distal PPG waveform [23]. The major drawback of the PTT and PAT approaches is the fact that two sensors are needed for measurement. This requires more effort and attention during setup since the obtained signals must be perfectly synchronised for accurate peak detection, which is vital in estimating these parameters. Additionally, the sensors are very sensitive to movements and in this case two signals require pre-processing, and analysis adding more complexity to the task.

In addition, several studies have proposed machine learning and deep learning models for cuffless BP estimation instead of using the traditional mathematical equations. These techniques are predominantly feature engineering-based approaches [24–28], while few other studies have attempted BP estimation using the raw ECG and PPG signals instead [29]. The established regression models vary between linear regression, random forest, support vector machine (SVM), AdaBoost regression, decision trees and neural networks such as feedforward neural network and long short-term memory (LSTM). For example, in Kachuee et al [24,25] several of these models were evaluated and compared on roughly a thousand subjects acquired from the multi-parameter intelligence monitoring in intensive care units (MIMIC II). PTT features along with PPG morphological features were extracted from the pre-processed ECG and PPG signals and the arterial blood pressure (ABP) signal was used as ground truth for SBP and DBP values. In their experiments, the Adaboost model achieved the best results with mean absolute error (MAE) and standard deviation (SD) of 11.17 ± 10.09 mmHg and 5.35 ± 6.14 mmHg for SBP and DBP respectively (using a calibration-free approach).

Classical machine learning models and non-recurrent neural network models face a major accuracy decay problem for long term BP estimation. This issue occurs due to their inability to model the temporal dependencies in the data which in turn requires frequent calibration. In order to overcome this challenge, Su et al [28] and Li et al [30] proposed a deep recurrent neural network equipped with LSTM units for cuffless BP estimation. Seven features extracted from the ECG and PPG were used as input vector to the deep learning model. The results in both studies outperform existing models in the literature and showed a good performance for long term BP estimation. In a recent trend, some authors have attempted to use a combination of raw ECG, PPG and ballistocardiograms (BCG) signals as input for BP estimation models. In Tanveer and Hasan [29], the raw ECG and PPG signals of 16 and 40-seconds window have been evaluated on an ANN-LSTM network using 39 subjects. The MAE of the proposed method were 0.93 mmHg and 0.52 mmHg for SBP and DBP, respectively, on the 40 s window. In another study, Eom et al [31] used the raw PPG, ECG and BCG signals for estimating BP using a combination of convolutional and bidirectional GRU network. The model was evaluated on a relatively small dataset consisting of 15 subjects only. The study reported a MAE of 4.06 ± 4.04 mmHg for SBP and 3.33 ± 3.42 mmHg for DBP. Although these studies do not require feature extraction, they still share the disadvantages associated with the PTT/PAT approach as mentioned earlier. Overall, this approach has been studied thoroughly in the past decades, and a variety of different methods were proposed, tested and compared, with some reporting notable performance.

Most recently, the rise of machine learning and advances in wearable technologies have given way for new research towards cuffless BP estimation using just a single PPG sensor. A review paper by El Hajj and Kyriacou has comprehensively covered the available literature on BP estimation using machine learning [32]. The PPG has been shown to relate directly to volumetric changes of blood within an artery. The deterioration of the artery’s elastic properties either due to disease or age will have an effect on blood pressure [33]. These changes will also be reflected on the morphology of the PPG signal, and hence, the PPG could be considered as a suitable signal for the non-invasive estimation of blood pressure [34,35]. Many research studies have attempted to estimate BP by focusing on the analysis of the PPG morphology features that generally capture the shape of the PPG waveform. The PPG signal can give insights into the state of the cardiovascular system and has been used for decades in medical devices for measuring heartrate and oxygen saturation [34]. Photoplethysmography is a simple, low cost and non-invasive optical method used for measuring the volumetric changes in blood. This is done by illuminating the skin with a light-emitting diode (LED) and measuring the amount of absorbed or reflected light from the vasculature, over time, using a photodetector. The amount of reflected light is governed by the blood circulation in the illuminated skin. When the heart contracts during systole, the BP increases as a result of the blood pulse propagating from the heart towards the periphery. On the other hand, the BP decreases during heart diastole. The PPG signal is synchronous to the blood pulse and has a wavelike form. Consequently, the repeated systole and diastole of the heart result in a periodic PPG signal exhibiting one systolic and one diastolic peak in every single cardiac cycle. The turning point or the valley between the systolic and diastolic peaks is called dicrotic notch. A typical PPG cycle is shown in Fig. 1. The systolic peak is normally clear and easy to detect while the diastolic peak and dicrotic notch are normally visible mostly in healthy and young people and become undetectable in older and hypertensive patients. Although the relationship between the PPG and BP is not yet clear, the small variation in blood volume reflected in the PPG signal appears to be correlated with BP [34].

Currently, there is no mathematical equation to model the relationship between the PPG and BP. However, several studies have established this relationship using machine learning and deep learning models. Most of these studies rely mainly on time domain features extracted from the PPG morphology and its derivates, such as amplitudes, width, area under the curve, infliction points, etc. For example, Teng and Zhang [10] investigated the relationship between BP and few features extracted from the PPG waveform. The study was evaluated on 15 healthy subjects using a linear regression model. The reported mean error for SBP was 0.21 ± 7.32 mmHg and 0.02 ± 4.39 mmHg for DBP. Kurylyak et al [11] estimated BP using 21 PPG morphology features extracted from an

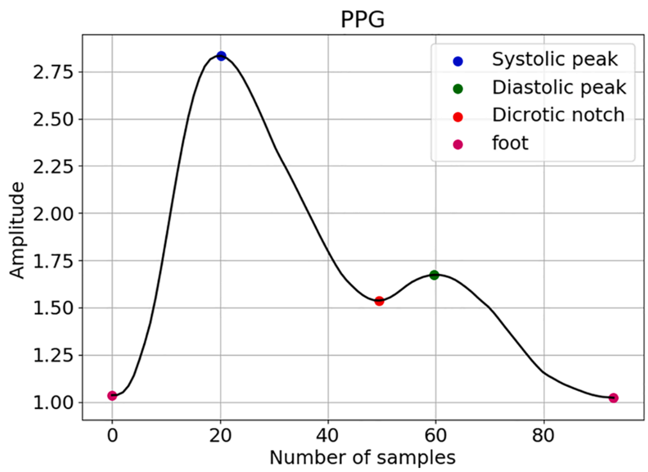

Fig. 1. Typical PPG cycle with key points.

unspecified number of patients acquired from the MIMIC II dataset. The relationship was modelled using a feedforward neural network. The MAE for SBP was 3.8 ± 3.46 mmHg and 2.21 ± 2.09 mmHg for DBP. Liu et al [12] included an additional 14 features derived from the second derivate of the PPG. A support vector machine was used as a BP estimation model evaluated on a small number of PPG cycles. The 14 added features improved the model performance when compared with the performance on 21 PPG features only. Unlike all the previous studies that rely on temporal PPG features. Xing and Sun [14] extracted frequency domain features (amplitudes and phase features) from the PPG using a Fast Fourier Transformation. This study was conducted on 69 patients from the MIMIC II dataset and 23 volunteers. A feedforward neural network was applied for estimating BP. The results were comparable with other related work in the field. However, all the aforementioned models are not best fit for continuous BP estimation since they do not model the temporal dependencies in the PPG features and their accuracy will decrease rapidly over time.

In accordance with the recent trend of using raw signal instead of feature engineered approach, Sideris [36] et al evaluated the relationship between BP and raw PPG signals using an LSTM model. The dataset was derived from the MIMIC database with originally 200 patients, however, after pre-processing including removing inappropriate and short signals, only 42 patients were fit for the study. The advantage of this study is that the LSTM was able to estimate the continuous arterial blood pressure (ABP) with reasonable performance using only PPG signals. The root mean squared error for SBP was 2.58 ± 1.23 mmHg and 6.042 ± 3.26 mmHg for the continuous ABP. In another research study, Slapnicar et al [37] established a ResNet model for estimating BP using raw PPG and its first two derivatives without feature engineering. The model was evaluated on 510 subjects collected from the MIMIC III database. Both calibration-free and calibration-based approaches were assessed and compared in their study. The calibration-based approach performed better, with MAE of 9.43 mmHg for SBP and 6.88 mmHg for the DBP, hence the overall model performance was poor. In a latest study published by Sadrawi et al [38], the authors attempted to estimate the ABP waveform using a genetic based deep-convolutional autoencoder (GDCAE). The model was evaluated on signals collected during surgical operation from 18 subjects. The study reported high correlation between the estimated and target SBP and DBP values. While the model demonstrated reasonable MAE with 2.54 mmHg and 1.48 mmHg for SBP and DBP, respectively, it was only evaluated on a very small number of patients.

Although some studies have shown remarkable performances using the PPG approach, this was mostly achieved using a relatively small number of patients. Uncertainty still remains around the accuracy of the PPG approach which is evident given the wide variation in the reported results. For this reason, the work presented here, underpinned by previous research on this subject [39], rigorously evaluates the capability of recurrent neural network models in predicting SBP and DBP, on a much larger patient subset from the MIMIC II database and with an increased number of input features. The main contributions of this study are summarised as follows:

- Feature exploration and selection from the PPG waveform including its first (PPG') and second (PPG'') derivates
- Evaluate the proposed architecture on roughly a thousand subjects from the MIMIC II database
- The BP estimation models presented here achieved better results than previous similar studies

2. Methodology

The BP deep learning estimation algorithms employed in this study consist of a one bidirectional recurrent neural network (RNN) layer at the bottom (first layer), followed by several RNN layers and an attention layer. The RNN units utilised in this work are LSTM units and Gated Recurrent Units (GRU). Both unit types have been evaluated separately using the deep learning architecture. The input feature vector contains 52 features extracted from the PPG, PPG' and PPG''. The feature vector was then reduced using machine learning dimensionality methods in order to reduce collinearity and remove redundant features that negatively affect the estimation precision and increase the chances of overfitting.

2.1. Dataset

The data used for evaluating the BP estimation algorithms were derived from the MIMIC II dataset [40]. This dataset is provided online by Physionet and contains several waveforms measured simultaneously from thousands of patients in intensive care units (ICU). The dataset includes signals such as PPG recorded from the fingertip, invasive ABP, ECG, etc. Both PPG and ABP signals are sampled at 125 Hz. The MIMIC dataset provides a variety of subjects from different age groups and gender with potentially varying range of BP values, which are of great value for this study.

The original raw signals acquired from the MIMIC II are noisy and sometimes corrupt and require a lot of pre-processing prior to their utilisation in the analysis. Fortunately, Kachuee et al [24] have published clean and pre-processed data from the MIMIC II and hence, we adopted their version of the MIMIC II dataset that are currently presented in the University of California, Irvine (UCI) Machine Learning Repository. The records in this dataset are extracted from 942 patients. Kachuee et al stored the data in a very efficient and convenient format for analysis. The dataset consists of three signals, PPG, ECG and ABP stored in a cell array of matrices where each cell corresponds to one record. Each row in each matrix represents one signal channel. For this study only the PPG and the reference ABP signals were used.

For our data analysis, the signals were segmented into 10 s frames in order to take full advantage of the LSTM and GRU structures in processing sequential data for modelling the time domain variation in the PPG features. The ground truth SBP and DBP values were extracted from the ABP signal, which was not pre-processed nor filtered in order to keep the peaks and valleys as accurate as possible. The SBP and DBP correspond to the average of the detected peaks and valleys, respectively, in the 10 s sequence. The peak is the highest value in each single cardiac cycle whereas the valley is the end-diastole value. In our experiments, every single cycle must have one SBP and one DBP value extracted from it, otherwise the cycle is considered invalid, and hence ignored. Furthermore, we excluded sequences that correspond to very low or very high SBP and DBP values (e.g. SBP $\geq$ 180, DBP $\geq$ 130, SBP $\leq$ 80, DBP $\leq$ 60). Moreover, given the large size of the dataset presented by Kachuee et al [24] (more than 740 h in total) and in anticipation of the computational complexity of the deep learning models, we further reduced the size of the data in an effort to speed up the training and alleviate the computational requirements. Table 1 and Fig. 2 present the statistical information about the ranges of 10-s averaged SBP and DBP values in terms of min, max, mean and standard deviation in the dataset.

2.2. Data pre-processing

Since the dataset utilised in this study was compiled and published by Kachuee et al [24], the filtering and denoising step is the identical to the method presented in their paper. Several pre-processing techniques were tested in their study, and after analysis the discrete wavelet decomposition (DWT) was selected given its performance advantages over the others in terms of complexity, phase response and handling of different levels of noise in the signals. The DWT was applied to the signals performing ten decomposition levels, with Daubechies 8 (db8) set as the mother wavelet. Afterwards, the decomposition coefficients corresponding to the very low (0 to 0.25 Hz) and high (250 to 500 Hz) frequency components were set to zero in order to eliminate them. The wavelet denoising was then performed with soft Rigrsure thresholding on the rest of the decomposition coefficients. The pre-processed PPGs free from low frequency baseline fluctuation were obtained by reconstructing the decomposition.

In order to make the feature extraction process more accurate and robust, the PPG's amplitude was normalised to a range of [0,1]. The amplitude of the PPG varies greatly between subjects and this introduces two challenges. First, the arbitrary change in the PPG amplitude makes the cycle per cycle segmentation (for feature extraction) more difficult since it depends on a fixed PPG peak detection threshold. This in turn may cause cycles with lower PPG peaks than the peak threshold to go undetected and, hence losing valuable data and as a result reduce the BP range in the final dataset. Second, some of the extracted features may depend on the scale or amplitude of the PPG and its derivatives. Therefore, all PPG signals were normalised in order to make the feature extraction process more efficient and ensure that the extracted feature values are meaningful. The PPG signals were normalised using the min-max normalisation method.

2.3. Features

As mentioned earlier, in the PPG approach, BP is commonly estimated using a set of features computed from the PPG contour describing its morphology on a per-cycle basis i.e. features extracted from every cardiac cycle. Hence, the estimation accuracy of this approach relies heavily on the quality of the features, coupled with a high-performing prediction model.

In this study, we expand on the 22 PPG feature set defined in a previous study [39] by adding more features that describe the PPG waveform as well as features extracted from the first (PPG') and second (PPG'') derivatives. PPG extracted features include width at different amplitudes, heartrate, systolic upstroke time, diastolic time, pulse intervals, area under the curve, etc as well as various other features from its derivatives. Those features are frequently used in the BP related work in the literature. A typical PPG single segment, in an ideal scenario, contains two foot (one at each end) with a systolic peak, diastolic peak and dicrotic notch in between, as shown in Fig. 1. In this particular scenario the PPG' also has two peaks with one positive to negative zero-crossing after each peak. However, in our dataset, given the fact that the PPGs are collect from ICU patients (i.e. under medication which could influence the BP and the PPG waveforms, as well as other factors such as diseases and age) some these distinctive features may not be dominant or visible, particularly the dicrotic notch and diastolic peak. This is also reflected in the PPG' and PPG'' making it very difficult to compute related features such as delta T (time between systolic and diastolic peaks), dicrotic notch time, augmentation index, large artery stiffness index and many other important features. Therefore, all features related to dicrotic notch and diastolic peak were excluded in our study. It is essential however for our analysis to have two clear foot and one systolic peak since all extracted features from the PPG, PPG' and PPG'' depend on them. In total, we computed a 52-feature set presented in Table 2 and shown in Fig. 3.

Several of the features computed here have already been linked or proven to have influence on BP. It has been suggested that the systolic peak is rather more suitable for continuous BP estimation than PAT [41]. In another investigation published by Awad et al [42], pulse width at 50 % amplitude has been found to be correlated with total peripheral resistance or systemic vascular resistance. Pulse area is another indicator of total peripheral resistance as reported in a previously published study by wang et al [43]. Kurylyak et al [11] expanded on the findings of Awad et al study by considering a total of 21 width related features from the systolic and diastolic phases and their ratios at 10 %, 25 %, 33 %, 50 %, 66 % and 75 % amplitude of the PPG, as well as systolic time, diastolic time and cardiac period. Their study aimed to include as many PPG features as possible in order to capture the shape of the waveform accurately and improve the estimation performance. It has been suggested by Poon et al [44] that the ratio of pulse interval to its peak amplitude provides an understanding of the properties of the cardiovascular system of a person. Systolic upstroke time, referring to the time between the start of the PPG cycle and the systolic peak point reflects changes in BP and can be used for cardiovascular disease classification [45]. PPG intensity ratio (PIR) has been used in several studies as a potential indicator for BP estimation [27,46]. PIR is correlated with the arterial diameter change, which has an impact on the total peripheral resistance [46]. Moreover, height ratios of the PPG'' extracted from the a-wave, b-wave and e-wave denoted as A_b/A_a and A_e/A_a are linked to arterial distensibility, and age. For example, higher A_b/A_a ratio reflects an increased arterial stiffness which increases with age, while higher A_e/A_a ratio indicates decreased arterial stiffness hence decreases with age [41]. Both arterial distensibility and total peripheral resistance are direct influencers of BP.

2.4. Feature selection

Feature selection or dimensionality reduction is an essential step that aims to reduce the complexity of the model by eliminating redundant or irrelevant features which in turn helps to avoid the risk of overfitting the machine learning algorithms. The outcome of this process is a reduced set of features with the highest impact and influence on the BP estimation. In order to achieve this objective, several steps are involved in this process. First, the extracted feature values vary greatly in scale, for example, systolic upstroke time has incredibly small values compared to the heart rate values. Hence, these features are not comparable, which leads to imbalanced analysis. Furthermore, the estimation models favour larger feature values, therefore, less attention is placed on smaller values in the input vector, and as a result it weakens and undermines their actual influence on BP. This impacts the performance and accuracy of the BP estimation models. Therefore, for fair analysis and comparisons to be made between the extracted features and their relationship to BP, all features were normalised to a range of [0,1]. Another reason for normalising the input feature vector is to reduce the effect of outliers in the dataset and in turn boost the performance of the BP estimation algorithms. The features were normalised using the following equation:

$$x^{'} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Table 1
Statistics for the SBP and DBP values in the final dataset.

	Min (mmHg)	Max (mmHg)	Mean (mmHg)	SD (mmHg)
SBP	80.09	179.99	134.30	19.80
DBP	60.00	129.58	73.48	10.04

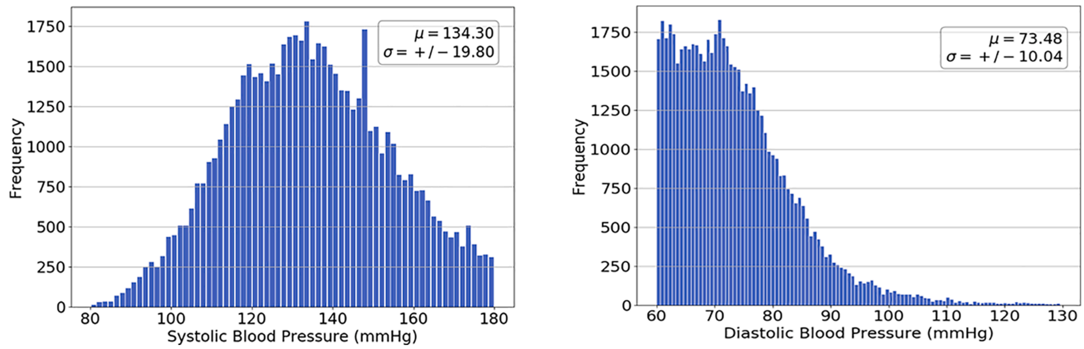

Fig. 2. SBP and DBP ranges in the dataset.

In the second step, we applied methods to assess the correlation among the extracted features and their relationship to BP. The methods involved in this process are namely Pearson's correlation coefficient and maximum information coefficient (MIC) [47]. The former is used to evaluate the linear relationship between each pair of features, while the latter is used to evaluate the nonlinear relationship between the features and BP. For our analysis, Pearson's correlation was applied to identify each pair of features with high linear correlation in an attempt to reduce collinearity in the input vector. Collinearity introduces redundancy in the data, and this increases the computational complexity (more space and time needed) and the risk of overfitting the model. In this study, we considered two features to be highly correlated if their Pearson's correlation coefficient is bigger than or equal to 0.9. Afterwards, we analysed the results of the MIC between all features and BP. MIC values were between 0 and 1, and similar to Pearson's correlation coefficient, a higher value means higher correlation/dependency between the pair. Therefore, among each pair of colinear features, we dropped the feature with the smaller MIC value against BP, while the other one was retained for further analysis.

Finally, in the last step, we applied recursive feature elimination (RFE) [48] on the feature vector obtained from the previous step. RFE is a popular wrapper-type feature selection method. Wrapper methods incorporate a machine learning algorithm, used in its core, during the feature selection process. RFE starts with all feature-set and iteratively removes features until the optimum or specified number of features is reached. RFE works by fitting a learner model, ranks the features by importance based on the results of the regression model, eliminates the features with the least contribution on the target estimation and refits the model. This process is repeated until the optimum number of features is achieved. The learner model wrapped in the core of RFE in this study was random forest. The RFE has two configurations, either the user specifies the number of desired features or the algorithm finds the optimal number of features. We chose the latter, and as a result, the RFE selected 22 features for DBP and 23 features for SBP as optimal number of predictors. After analysing the results, we found that most of the selected features were common for SBP and DBP but with different importance. These features are listed in Table 3. We combined all these features together to form the final input feature vector for the deep learning models. As a result, we were able to reduce the feature set by more than half, from 52 features down to 24 features only. Two datasets were created to accommodate both feature sets separately for evaluation using the models described in the following section.

2.5. Models

The BP estimation task using the PPG approach is attractive but still a fairly complicated task. Perhaps the main challenge associated with this approach is accuracy and this is due to the lack of clear understanding of the nature of the relationship between the PPG features and BP. However, recent research suggests that this relationship is inherently nonlinear [11,24,25]. This is apparent given that the majority of these studies achieved better accuracy with nonlinear functions, particularly neural networks. It is therefore clear that powerful deep learning models can potentially compensate for what we lack in terms of understanding of the exact relationship between BP and PPG features. In any case, the BP deep learning estimation model should account for the complexity of this relationship, as well as model the temporal variation in the extraction of PPG features in order to provide long term BP estimation whilst maintaining a good performance. In this study, the model architecture employed for this task took into account these challenges by providing non-linear processing units and the functionality to handle sequential time domain data. This architecture has been proposed in a previous study [39] and was shown to outperform the conventional feedforward neural network, LSTM and GRU. The model used is a recurrent neural network (RNN) model with bidirectional connections in the first layer, followed by a stack of *n* conventional recurrent layers (unidirectional) and an attention layer. Unlike the hyperbolic tangent (tanh) units in the conventional RNN, we used LSTM and GRU cells.

2.5.1. Multilinear regression

We established a multilinear regression (MLR) as a baseline machine learning model for comparative purpose only. MLR assesses the linear relationship between *n* independent input features and one dependent variable given the following equation:

$$Y = \beta + X_0\beta_0 + X_1\beta_1 + X_2\beta_2 + X_3\beta_3 \cdots X_n\beta_n + \varepsilon \quad (1)$$

Where Y is the estimated BP (SBP or DBP), $X_0, X_1, \ldots, X_n$ are the PPG input features, $\beta_0, \beta_1, \ldots, \beta_n$ are the model coefficients and ε is the random error (difference between the estimated and reference value). The coefficients were optimised using the mean square error (MSE).

2.5.2. LSTM and GRU cells

The LSTM is a special type of RNNs that was established to address the vanishing gradient problem [49]. This problem occurs when training a deep neural network or RNN with activation function such as sigmoid or tanh. Training a neural network involves computing the prediction error at the output layer and estimating a gradient. This gradient error is essentially used by the network to update or optimise the weights at each layer as it propagates backward from the final layer to input layer. However, with deep neural networks i.e. many layers, the gradient decreases dramatically as it moves backward through the layers and gets very small as it approaches the initial layers, this results in little or no update to their weights and biases. The LSTM resolves this issue by replacing the conventional RNN tanh units by a memory cell state C_t that preserves information from previous time steps using three multiplicative units that regulate the flow of information. This allows the LSTM to handle long term dependencies in sequential data more effectively than conventional non-recurrent neural networks and RNNs. The LSTM cell is shown in Fig. 4(left).

At time step t, the LSTM takes the current input X_t and the output

Table 2
This table presents features derived from the PPG, PPG' and PPG''

Feature	Definition
PPG − 34 features	
Systolic peak (x)	Peak amplitude of the waveform
Pulse width and ratios @:	
10 % amplitude	DW10; DW10 + SW10; DW10/SW10
25 % amplitude	DW25; DW25 + SW25; DW25/SW25
33 % amplitude	DW33; DW33 + SW33; DW33/SW33
50 % amplitude	DW50; DW50 + SW50; DW50/SW50
66 % amplitude	DW66; DW66 + SW66; DW66/SW66
75 % amplitude	DW75; DW75 + SW75; DW75/SW75
Systolic area (A1)	Area under curve between start and peak of the waveform
Diastolic area (A2)	Area under curve from the peak to the end of the waveform
Pulse area	A2/A1
Pulse interval (t_pi)	Distance between the start and end of the cycle
t_pi/x	Ratio of the pulse interval to its systolic peak
Heart rate (HR)	60/peaks interval
Systolic upstroke time (ST)	Time interval between the start and peak of the waveform
Diastolic time (DT)	Time interval between the peak and end of the waveform
ST/t_pi	Ratio of the systolic upstroke time to its pulse interval
DT/t_pi	Ratio of the diastolic time to its pulse interval
Main wave rising slope (slope)	Value of the PPG at the peak index of its first derivative
Slope/x	Relative height of the slope point
t_slope	Time interval between the slope point and the peak
t_slope/t_pi	Ratio of time interval between the slope point and the peak to its pulse interval
PPG intensity ratio (PIR)	Ratio of the peak point intensity to the foot (valley) intensity
PPG first derivative (PPG') – 5 features	
Peak amplitude (a1)	Intensity of the first maximum peak of the PPG'
Peak time (t_a1)	Time interval between the beginning and peak (a1)
Valley time (t_b1)	Time interval from the peak (a1) to first valley (b1)
t_a1/t_pi	Ratio of the PPG' peak time to its PPG pulse interval
t_b1/t_pi	Ratio of the PPG' first valley time to its PPG pulse interval
PPG second derivative (PPG'') – 13 features	
First peak amplitude (A_a)	Intensity of the first maximum peak (a-wave)
First valley (A_b)	Intensity of the first valley of the PPG'' after A_a
A_b/A_a	Ratio of the first valley intensity to the first peak intensity of the PPG''
AP_b/AP_a	Ratio of the PPG amplitude at the index of A_b to the amplitude of the PPG at the index of A_a
A_e/A_a	Ratio of the second peak intensity e-wave A_e (occurring after the first valley of the PPG'-b1) to the first peak of the PPG''
AP_e/AP_a	Ratio of the PPG amplitude at the index of A_e to the amplitude of the PPG at the index of A_a
t_a2	Time interval from the start of the PPG'' to its first peak A_a
t_b2	Time interval between A_a to the first valley A_b
t_c2	Time interval between the first valley A_b to the second peak A_e
Total PPG'' intensity	Total intensity of PPG'' between peak A_a and valley A_b
t_a2/t_pi	Ratio of t_a2 to the pulse interval of the PPG waveform
t_b2/t_pi	Ratio of t_b2 to the pulse interval of the PPG waveform
t_c2/t_pi	Ratio of t_c2 to the pulse interval of the PPG waveform

from the previous hidden state h_{t-1} and pass it through a sigmoid function 'σ' called the forget gate f_t. The sigmoid function outputs values between 0 and 1 for each element of the previous cell state C_{t-1}. The information is kept as the output of f_t gets closer to 1 and removed as the output gets closer to 0. In the next step, the input gate i_t and the new candidate values $\hat{c}_t$ (output of the tanh layer) are multiplied together to create an update to the C_t. The input gate decides which values are kept from the $\hat{c}_t$. As illustrated by Fig. 4(left), in order to compute the C_t, we multiply C_{t-1} by the forget gate and then we add it to $i_t * \hat{c}_t$. Finally, the output gate o_t determines what information should carry to the next hidden state. We calculate h_t by multiplying the output of o_t by tanh (C_t).

Another variant of the RNN that has been more recently proposed to solve the long-term dependency issue is the GRU [50], illustrated in Fig. 4(right). Similar to the LSTM, the GRU regulates the flow of information using multiples gates embedded in its cell and provides superior performance in comparison to conventional RNNs. However, the GRU has a few advantages over the LSTM. Firstly, the GRU has only two gates, namely, an update gate and a rest gate. Secondly, the GRU does not have a memory cell and uses the hidden state to pass the information instead. Therefore, the GRU is computationally more efficient in terms of required space and time as it has less parameters, hence it trains fasters than LSTMs. The GRU works in a similar fashion as the LSTM, where at each time step the GRU cell has two inputs which include the current input X_t and the output of the pervious hidden state h_{t-1}. The r_t and z_t are the rest and update gates, respectively. The rest gate decides how much information to forget from the previous hidden state output h_{t-1}. The update gate combines the role of the input and forget gates of the LSTM which decides what information to discard and what to add to the new hidden state h_t.

2.5.3. Bidirectional recurrent layer

Conventional LSTM and GRU are unidirectional with forward layers only. As described in the previous section, their hidden state cell incorporates only past history i.e. h_{t-1} in estimating the current output h_t. In contrast, in the bidirectional layer two hidden layers are created, and their outputs are concatenated into one output layer [51]. Each bidirectional layer has a forward and a backward layer. The forward layer allows the hidden state to consider past information, while the backward layer allows the hidden state to consider near future information. Thus, having two layers expose the hidden state to more information and in turn this allows the network to improve its ability in handling long-term dependencies and increases its performance. A bidirectional layer containing a forward layer and backward layer is shown in Fig. 5. On one hand, the forward layer processes the input in a normal time order, and the forward hidden vector $\overrightarrow{h_t}$ is calculated in a similar way as described earlier in the conventional unidirectional LSTM and GRU layer. On the other hand, the backward hidden vector $\overleftarrow{h_t}$ is obtained by processing the input in reverse order. The final output of the bidirectional hidden vector h_t is a concatenation of the forward hidden sequence $\overrightarrow{h_t}$ and the backward hidden sequence $\overleftarrow{h_t}$.

2.5.4. Attention mechanism

Attention mechanism has been successfully applied in neural machine translation [52] and computer vision [53]. The main purpose of the attention mechanism is to allow the network to focus on the most relevant parts of a sequence that contribute more to the target output. In the context of the proposed model, the feature vectors or hidden states may have varying influence on the BP estimation, and the attention mechanism will train the model to automatically identify the feature vectors with higher contribution on the target output in each time step. Therefore, features carrying more important information are assigned larger weights. In this study, we applied a self-attention model which implements a single feedforward layer with tanh activation function to approximate a score for each hidden state vector in every time step [52]. This score is a measure of correlation or importance between the input and the target. Afterward, we passed the obtained attention weights vector to a softmax function to normalise its values to a range between [0,1]. The final output vector V is generated by computing the weighted sum of the attention weight vector α and their hidden state vector h, as presented in the following equation:

$$V = \sum_{i}^{n} \alpha_i h_i$$

2.5.5. Model architecture

The architecture of the deep learning models, used in this study, consist of one bidirectional RNN layer (Bi-RNN), followed by a stack of n

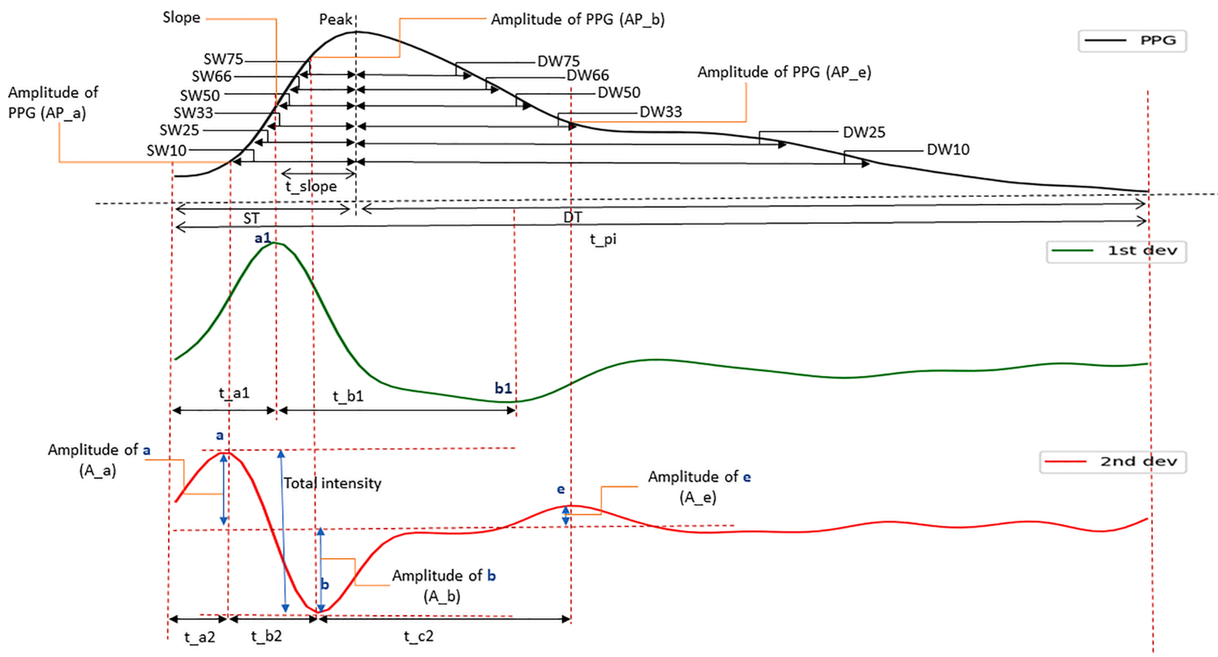

Fig. 3. Illustration of temporal features from the PPG and its derivatives.

Table 3
Features selected by the RFE for SBP and DBP listed by order of importance.

# of features	SBP	DBP	Combined
1	A_e/A_a	A_e/A_a	A_e/A_a
2	t_b1	DW25/SW25	DW25/SW25
3	DT	DT	DT
4	DW25	t_slope/t_pi	t_slope/t_pi
5	t_b2	A_b	A_b
6	t_a1/t_pi	t_a1/t_pi	t_a1/t_pi
7	t_b1/t_pi	DW25	DW25
8	DW66/SW66	t_c2	t_c2
9	DW10	DW10	DW10
10	t_c2	DW50/SW50	DW50/SW50
11	t_slope/t_pi	t_b2	t_b2
12	A_b/A_a	t_a1	t_a1
13	DW50/SW50	t_b1/t_pi	t_b1/t_pi
14	DW25/SW25	t_b1	t_b1
15	ST	A_b/A_a	A_b/A_a
16	t_c2/t_pi	DW75 + SW75	DW75 + SW75
17	DW75 + SW75	DW50	DW50
18	DW10/SW10	PPG peak (x)	t_pi/x
19	t_a1	t_c2/t_pi	t_c2/t_pi
20	DW50	DW66/SW66	DW66/SW66
21	t_b2/t_pi	ST	ST
22	A_b	DW10/SW10	DW10/SW10
23	t_pi/x		t_b2/t_pi
24			PPG peak (x)

unidirectional RNN (uni-RNN) layers and one attention layer. The bidirectional layer helps the network to capture more information by processing the input sequence in a forward and backward order. Additionally, we replaced the conventional RNN units by LSTM and GRU cells to solve the gradient vanishing issue that occurs during the training of a deep neural network. We also attempted to further improve the performance of the model by training the network to focus on the hidden states with significant information using attention mechanism. The SBP and DBP were then estimated using a linear activation function in the output layer. Two deep learning models were evaluated in this study, one is equipped with LSTM cells while the other is equipped with GRU cells. The number of units and the number of stacked unidirectional layers were optimised during training.

2.5.6. Training setting

The datasets were divided into 70% train, 15% validation and 15% test sets. The test set was reserved for the final evaluation of the optimised model and remained completely disjoint from the training data. The model selection criterion was based on the lowest validation error. The cost function used to optimise the models was the mean squared error (MSE) and Adam optimiser was selected for finetuning the models' parameters. The number of epochs and batch size were set to 300 and 128, respectively.

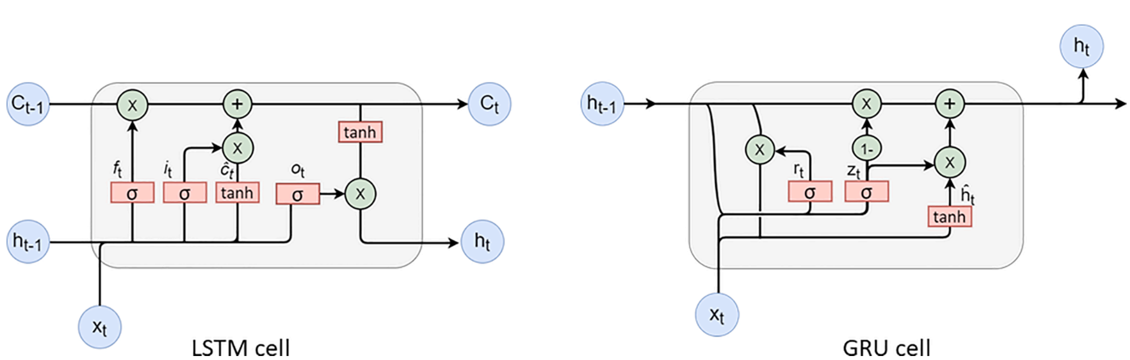

Fig. 4. LSTM (left) and GRU (right) internal cell structures.

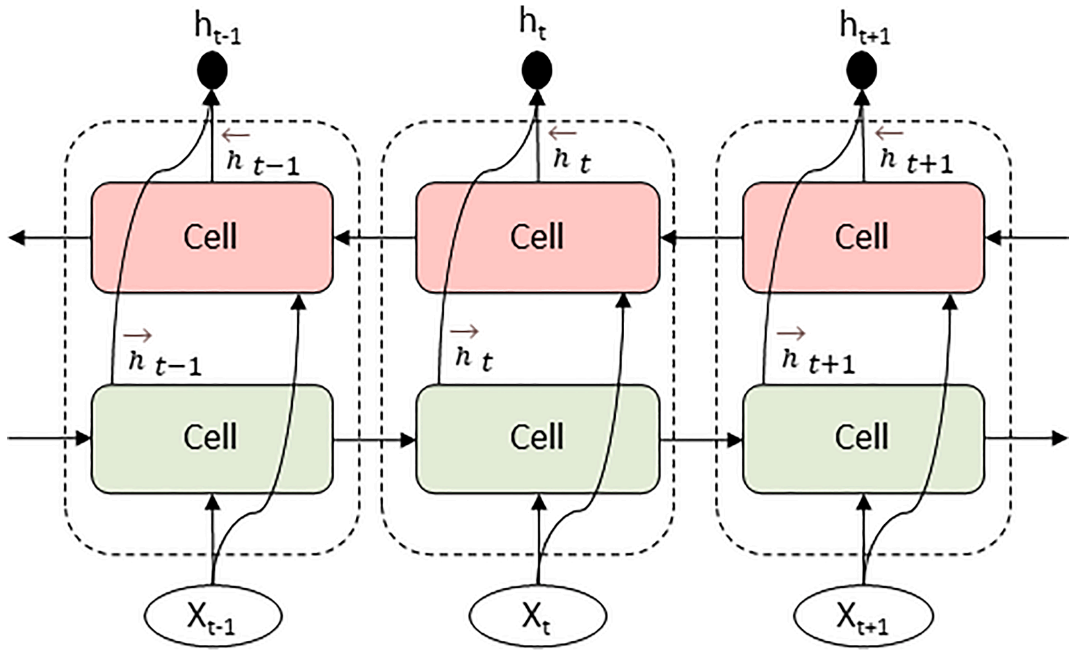

Fig. 5. Example architecture of a bidirectional layer with three-time steps. The conventional forward layer (green) and backward layer (red) are concatenated in each time step (black dot) to form the final output of a bidirectional layer.

3. Results

In this section, we evaluate and compare the previously discussed models and datasets. For simplicity, the BiLSTM and BiGRU refer to the bidirectional layer with LSTM units and GRU, respectively. The evaluation metrics used for assessing the performance of the three models were the mean absolute error (MAE) and standard deviation (SD). Additionally, we present further estimation analysis for the best performing model in each dataset using correlation analysis, and Bland-Altman plots.

3.1. Performance on 52 feature set

Table 4 presents the summary performance of the three evaluated models on the full feature set. The results show that the performance of the MLR is very poor compared to the deep learning models. The MAE for both SBP and DBP using the MLR model is roughly three times higher than the other two models. Overall, it can be said that the MLR is unable to estimate BP accurately, and there is a strong nonlinear relationship between the PPG features and BP. In comparison, the deep learning models outperformed the MLR by a huge margin. Both models were able to estimate BP with a good accuracy, particularly for the DBP. The results suggest that deep learning recurrent models are very promising toward cuffless and continuous BP estimation. Moreover, between the two deep learning models, the architecture equipped with LSTM cells was able to provide slightly better MAE estimation for both SBP and DBP. The best results (marked in bold in Table 4) were achieved using one BiLSTM layer, followed by 4 unidirectional LSTM layers and one attention layer using 512 hidden units in each layer. The MAE and SD for this model is 4.51 $\pm$ 7.81 mmHg for SBP and 2.6 $\pm$ 4.41 mmHg for DBP. This demonstrates that, to some extent, the relationship between the PPG and BP can be modelled with good accuracy.

Additionally, plots (a) and (b) in Fig. 6, show the regression plots and Pearson's correlation coefficient between the estimated and reference values for SBP and DBP for the best performing model. The evaluation indicates that the model predictions were consistent with the reference values with a very high correlation, particularly for the SBP. The correlation coefficient R for the SBP is 0.89 and 0.86 for the DBP, which indicates a very strong positive correlation between the ground truth and the prediction values. Furthermore, the Bland-Altman plots (c) and (d) in Fig. 6, demonstrate that the model was able to estimate BP over the whole range of SBP and DBP values that occur in the dataset. Moreover, the mean error between the reference and target were -0.48 mmHg and -0.49 mmHg for SBP and DBP, respectively. The lower and upper limits of agreement, represented by the two dotted lines, were [-18.42,17.45] for SBP and [-10.5,9.52] for DBP. This means that 95 % of error falls within these limits of agreements.

3.2. Performance on 24 feature set

The performance of the models evaluated on the reduced feature set is presented in Table 5. It can be seen that there is an insignificant change in the overall performance of the models on the 24-feature set compared to the performance on the full 52 feature set presented in Table 4. As expected, the estimation error of the MLR is much higher compared to all other models, especially for the SBP estimation.

Table 4
Results of various models on the 52-feature set.

Models	# of units, # of uni-RNN layers, & learning rate (LR)	52 features			
		SBP		DBP	
		MAE (mmHg)	SD (mmHg)	MAE (mmHg)	SD (mmHg)
MLR	–	14.86	10.88	7.14	6.3
BiLSTM + LSTM + attention	**(512, 4) & LR: 0.0001**	**4.51**	**7.81**	**2.6**	**4.41**
BiGRU + GRU + attention	(512, 1) & LR: 0.001	4.69	7.76	2.68	4.39

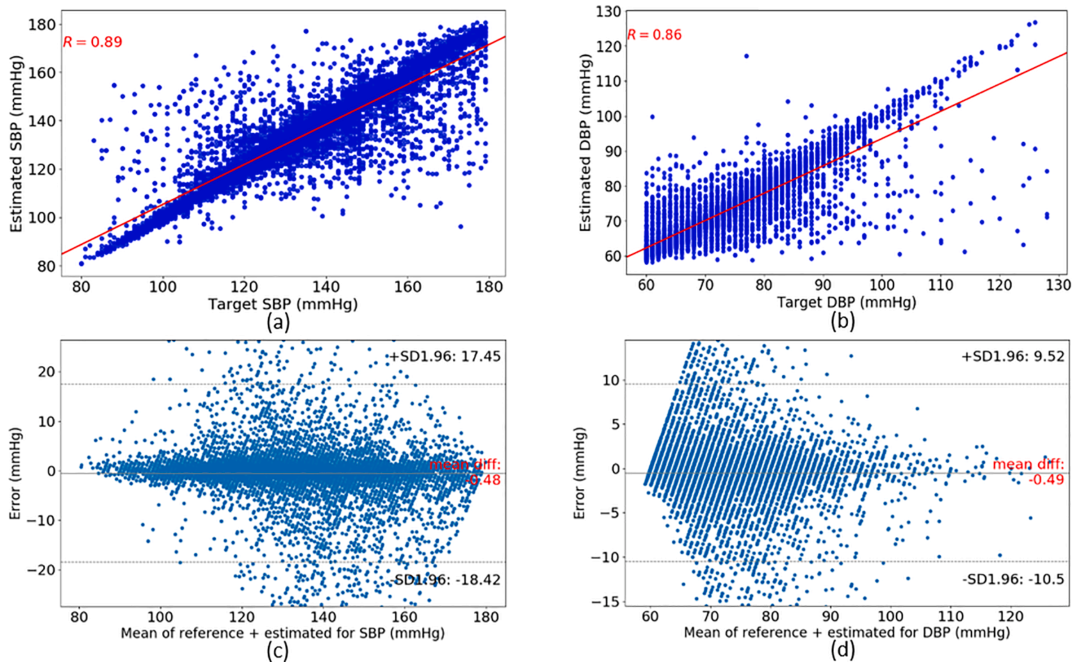

Fig. 6. Regression plots (top row) and Bland-Altman plots (bottom row) for the estimated SBP and DBP targets on the 52-features set.

Table 5
Performance of the three models on the 24-feature set.

Models	# of units, # of uni-RNN layers, & learning rate (LR)	24 features			
		SBP		DBP	
		MAE (mmHg)	SD (mmHg)	MAE (mmHg)	SD (mmHg)
MLR	–	15.11	10.95	7.42	6.6
BiLSTM + LSTM + attention	(512,4) & LR: 0.0001	4.86	8.43	2.83	4.86
BiGRU + GRU + attention	**(512,2) & LR: 0.001**	**4.79**	**8.08**	**2.77**	**4.72**

Therefore, the MLR failed to capture the relationship between the PPG features and BP. Similar to the performance from the previous dataset, the deep learning models produced acceptable results and their performance was superior compared to the MLR. Additionally, Table 5 shows that the deep learning models achieved roughly the same estimation accuracy for both SBP and DBP. The best performance based on the MAE and SD was achieved by one BiGRU layer followed by two unidirectional GRU layers and one attention layer using 512 units in each layer (marked in bold in Table 5). The MAE and SD is 4.79 $\pm$ 8.08 mmHg for SBP and 2.77 $\pm$ 4.72 mmHg for the DBP. The SBP and DBP estimation accuracy on this dataset was slightly lower compared to the previous results on the 52-feature vector. However, using the 24-feature vector and GRU cells, the computational complexity was significantly reduced, and therefore, the training was faster.

The regression plots and Pearson's correlation coefficients between the reference and estimated for both SBP and DBP again showed a very good correlation in the outcomes, as shown in plots (a) and (b) in Fig. 7. The correlation coefficient R for SBP is 0.88 and 0.84 for the DBP. Compared with the correlation analysis on the 52-feature set, we were able to achieve comparable outcomes, with negligible difference, using only 24 features. Similarly, the Bland-Altman plots (c) and (d) in Fig. 7, are within a close range in terms on mean error and limits of agreements as reported on the 52-feature set. The mean error for the SBP is -0.91 mmHg and -0.44 mmHg for the DBP. At 95 % confidence interval, the limits of agreement for the SBP were [-19.22, 17.41] and [-11.13, 10.25] for the DBP. Additionally, the model was able to estimate the low and high values of the SBP and DBP available in the dataset.

4. Discussion

In this study, we established three machine learning models for estimating SBP and DBP using only PPG signals. The MLR was used as a baseline model for comparative purpose only. The other models are deep learning models consisting of one Bi-RNN layer, followed by a series of stack uni-RNN layers and one attention layer. For these models, we conducted experiments using the LSTM and GRU cells instead of the tanh activation function. All models were evaluated on 942 subjects from the MIMIC II dataset. We extracted a total of 52 features from the PPG, and its first and second derivatives. The input feature vector was then reduced to 24 features using several feature selection techniques. The performances on the two datasets (52 vs 24 input features) were comparable, however, the models yielded slightly higher variance using the 24 features. This might suggest that adding redundant features or bias to the model could reduce the variance. In terms of accuracy, the results generated by the deep learning models were superior in comparison to the MLR. The MAE and SD for the DBP was much better compared to the SBP. This is due to the fact that the SBP had a considerably higher range of values, compared to the DBP range, which resulted in a higher estimation bias and variance. The best results were produced by the deep learning model with LSTM units using the 52 features. The main advantage of these deep learning models is their ability to learn temporal dependencies in successive PPG cycles during the mapping between the input and target, which improves the BP estimation. Another advantage is their capability to consistently provide good performance whilst avoiding the vanishing gradient problem in deep models. Lastly, the models were trained to attend to the hidden

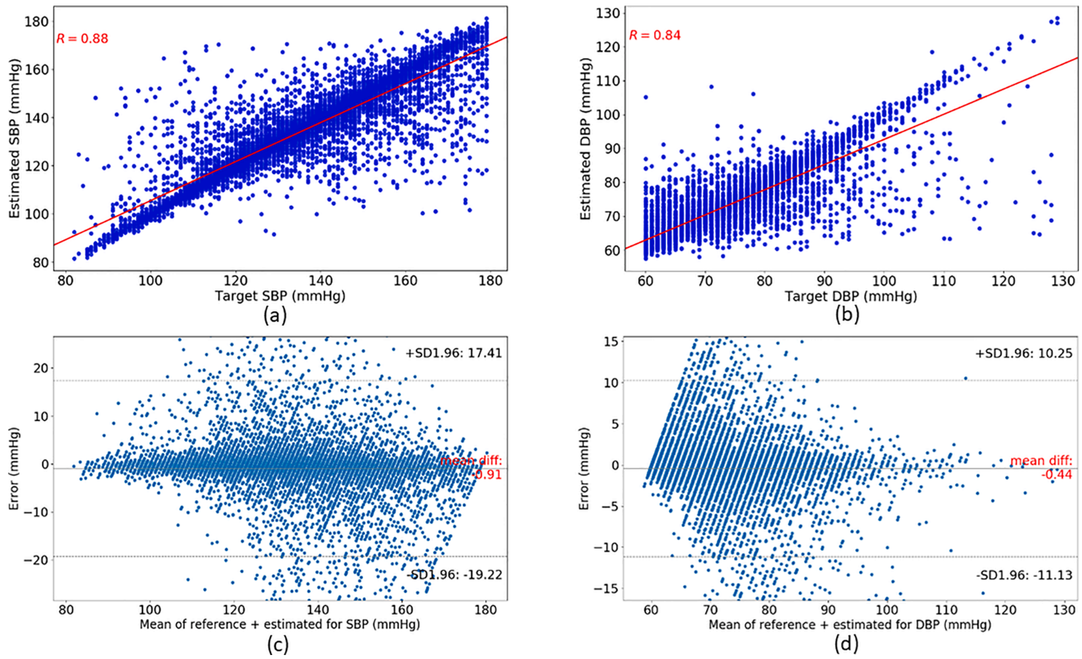

Fig. 7. Regression plots (top row) and Bland-Altman plots (bottom row) for the estimated SBP and DBP targets on the 24-features set.

Table 6
Performance evaluation of the best performing models on both datasets against the AAMI standards.

		Subjects	ME (mmHg)	SD (mmHg)
AAMI	SBP and DBP	≥ 85	≤ 5	≤ 8
Bi-LSTM + 4-LSTM + attention (Table 4)	SBP	942	−0.48	9.15
	DBP		−0.49	5.10
Bi-GRU + 2-GRU + attention (Table 5)	SBP		−0.91	9.34
	DBP		−0.44	5.45

states with the most significant influence on the BP, in each time step, using attention mechanism to enhance its estimation performance.

We assessed the performance of the best models (marked in bold in Table 4 and Table 5) against the international standards for cuffless BP monitoring set by the Association for the Advancement of Medical Instrumentation (AAMI) [54]. According to the AAMI criterion, the mean error (ME) and standard deviation (SD) of the model, evaluated on at least 85 subjects, should be less than or equal to 5 mmHg and 8 mmHg, respectively. The performance comparisons presented in Table 6, illustrate that the results comply with the AAMI. In particular, the ME and SD for the DBP were well below the AAMI margins, however, the SD for SBP on both datasets were slightly higher than the 8 mmHg limit. Nonetheless, the overall performance of these models confirms the feasibility of this approach.

In Table 7, we present a comparison between the best deep learning model (in Table 4) and some of the related works in the literature. It should be noted that we cannot make a direct and accurate comparison between our work and other related work since the data, acquisition devices and evaluation metrics are not consistent in all studies along with many other important information such as calibration. A large number of studies collect their own private data, while others use publicly available datasets with insufficient or unspecified number of subjects and potentially different subjects even when the same database is used. Furthermore, the evaluation metric also varies between studies, such as MAE, ME, root mean square error (RMSE), R-squared, etc. All these reasons make it very difficult to compare our work with other

Table 7
Performance comparisons with related works evaluated on the MIMIC database.

Authors	Dataset	Approach	Error (mmHg)
Kachuee et al [25]	851 subjects, (MIMIC II)	PTT feature-based, (SVR, calibration free)	MAE ± SD for SBP: 12.38 ± 16.17 DBP: 6.34 ± 8.45
Kachuee et al [24]	942 subjects (MIMIC II)	PAT feature-based, (Adaboost, calibration free)	MAE ± SD for SBP: 11.17 ± 10.09 DBP: 5.35 ± 6.14
Tanveer et al [29]	39 subjects (MIMIC I)	PTT raw signals, (ANN + LSTM, calibration free)	MAE for SBP: 0.93 DBP: 0.52
Kurylyak et al [11]	15,000 heartbeats (MIMIC II)	PPG features-based, (ANN, calibration free)	ME ± SD for SBP:3.8 ± 3.46 DBP: 2.21 ± 2.09
Slapnicar et al [37]	510 subjects (MIMIC III)	PPG + derivatives raw signals, (ResNet, calibration free)	MAE for SBP: 15.41 DBP: 12.38
This work	942 subjects, (MIMIC II)	PPG feature-based, (BiLSTM + LSTM + attention, calibration free)	MAE ± SD for SBP: 4.51 ± 7.81 DBP: 2.6 ± 4.41

studies. It was observed that, on one hand, reasonable estimation accuracy was achieved using a small dataset or limited number of patients. On the other hand, the performance worsens as the number of patients or the size of the data increase. Therefore, we will only compare our results with well-established studies (highly cited) that used the MIMIC public dataset, calibration free approach, and where the MAE or ME is used as an evaluation metric.

In Kachuee et al [24,25], several classical machine learning models were employed to estimate BP using PTT or PAT related features along with PPG features. The models were evaluated on a large number of subjects collected from the MIMIC II dataset. There are several limitations associated with this approach such as it requires two signals, often from two different measurement sensors, and both SVR and AdaBoost are not suitable for modelling the variation in the input features with respect to time, which generally enhances the prediction accuracy.

Furthermore, the reported MAE and SD for both SBP and DBP were high, as shown in Table 7. In comparison, our results were superior using only one PPG sensor. Tanveer et al [29] estimated BP using the raw ECG and PPG signals collected from 39 subjects from the MIMIC I. The BP estimation model used a feedforward layer to extract features from the input signals automatically, which were then fed into a LSTM model to account for the temporal variations in the data. This method achieved excellent results for both SBP and DBP. However, this approach still requires two signals, and the window length of the ECG and PPG was 40 s i.e. 5000 data points. Additionally, the accuracy also relied on the perfect synchronisation between the PPG and ECG signals. Furthermore, the model was evaluated on a small number of patients, which is insufficient. In our experiment, the model was evaluated on 942 subjects and we extracted features from a 10 s window of the PPG, hence the input feature vector was much smaller. In another study, Kurylyak et al [11], employed a feedforward model for estimating SBP and DBP using features extracted from the PPG. The results were reasonable in terms of ME and SD, however, the dataset was very small (15000 cycles), and the number of patients is unspecified. Slapnicar et al [37] proposed a ResNet based model for estimating SBP and DBP using the raw PPG including first and second derivatives signals as input. The model was evaluated on 510 subjects and the window length of each signal was 5 s. The network architecture is relatively complex, each of the three input signals are processed by five ResNet blocks and a spectro-temporal block with a GRU layer in parallel. This means that the complexity of the model is high. Additionally, the reported performance was poor compared to other reported studies in the literature. The MAE for SBP is 15.41 mmHg and 12.38 mmHg for DBP while the standard deviation was not reported. As shown in Table 7, in general, studies reported higher error and standard deviation on larger datasets. Overall, considering the large number of patients used in our study and the fact that we used only one signal, the model performance was relatively good. This again confirms that it is possible to effectively estimate BP using only one PPG signal with a reasonable accuracy, even when compared to the well-established PTT approach.

In terms of limitations associated with the PPG approach, these are mainly related to data availability and feature extraction. The data recordings in the MIMIC II are collected from ICU patients, and therefore, the subjects are generally sick, potentially older than the average population age, and under heavy medications. This means that the range of BP signals available in this dataset does not truly represent the general population. Additionally, the quality of the PPG signal is relatively poor. This, along with the fact that the PPG is collected from older and sick patients makes it impossible to extract all relevant features from the PPG morphology such as dicrotic notch and diastolic peak related features as well other features from the derivatives. Finally, the dataset available does not provide additional information about the patients such as height, weight, gender and age which can potentially help increase the estimation accuracy of the model.

5. Conclusion

In conclusion, in this study we explored a large number of features extracted from the PPG and its derivatives in an attempt to enhance the BP estimation. The deep learning estimation models employed here have an advantage over existing classical machine learning models and non-recurrent models in dealing with long term dependencies in the time domain data as well as improve the performance of conventional LSTM and GRU models. The models were evaluated on 942 patients extracted from the MIMIC II dataset. The obtained accuracies prove the feasibility of estimating BP without a cuff, using only one PPG sensor as opposed to two sensors using the PTT/PAT approach, which is costly, impractical, and needs additional requirements. Moreover, prediction analysis demonstrates that there is a very high positive correlation between the reference and estimated SBP and DBP values. The ME and SD for the DBP fulfil the AAMI requirement while the SBP results were within the required ME but exceeded the SD requirement by a very small margin. In future work, the deep learning models developed here can be evaluated on better quality data (e.g. non-ICU patients) and include demographical information.

CRediT authorship contribution statement

C El-Hajj: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Visualization, Investigation. **P.A Kyriacou:** Supervision, Writing - review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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